A Category-(c) Normalizable Frame: An Explicit Counterexample to Conjecture 3.14

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Status. Candidate counterexample, likely valid, pending expert review. An explicit normalizable frame in $\ell^2(\mathbb{N})$ belongs to category (c) of Yu's Theorem 3.13, disproving Conjecture 3.14.

Abstract

We construct a countable frame whose normalization is again a frame and whose successive norm bands are all frames, with summable optimal lower bounds and optimal upper bounds converging to zero. Every frame operator is diagonal in the standard basis and all optimal bounds are computed exactly. The construction gives a full negative answer to Conjecture 3.14 in Yu's *Frame-normalizable Sequences*.

1 Source problem

Let (x_n) be a sequence of nonzero vectors in a Hilbert space. It is *frame-normalizable* if the unit-vector sequence $(x_n/\|x_n\|)$ is a frame, and it is a *normalizable frame* if both (x_n) and its normalization are frames.

Yu's Theorem 3.13 divides normalizable frames into three categories. Its category (c) consists of those for which there are positive numbers $\delta_k \downarrow 0$ such that every norm band

$$\mathcal{F}_k = \{x_n : \delta_{k+1} \le \|x_n\| < \delta_k\}$$

is a frame. If A_k, B_k are the optimal lower and upper frame bounds of $\mathcal{F}_k$, the theorem also proves $(A_k) \in \ell^1$ and $B_k \to 0$.

Theorem 3.13. If $\{x_n\}$ is a normalizable frame for H, then it falls into exactly one of the following three categories.

(a) $\{x_n\}$ is a frame that is norm-bounded below.

(b) There exists a $\delta > 0$ such that $\{x_n \mid \|x_n\| \ge \delta\}$ is a frame and $\{x_n \mid \|x_n\| < \delta\}$ is not empty but does not satisfy a lower frame condition.

(c) There exists a sequence of positive scalars (δ_k) that decreases to 0 such that $\mathcal{F}_k = \{x_n \mid \delta_{k+1} \le \|x_n\| < \delta_k\}$ is a frame for every k. In particular, if A_k and B_k are optimal lower and upper frame bounds for $\mathcal{F}_k$, respectively, then $(A_k) \in \ell^1$ and $\lim_{k\to\infty} B_k = 0$.

Figure 1: The definition and necessary bounds for category (c) in Yu's Theorem 3.13 [1, p. 8].

Immediately after its proof, Yu proposes the following conjecture:

"There does not exist a normalizable frame belonging to category (c) in Theorem 3.13."

Based on our observations, we propose the following conjecture.

Conjecture 3.14. There does not exist a normalizable frame belonging to category (c) in Theorem 3.13. $\diamondsuit$

Figure 2: Conjecture 3.14 in the source paper [1, p. 9].

2 Proof intuition

A single norm band must be a frame for the whole Hilbert space even though all its vectors have one small common norm. The normalized version of that band can therefore have a very poor lower frame bound. We realize this with two vectors on each of infinitely many orthogonal coordinate pairs. Their opposite signs cancel all off-diagonal terms, leaving a block frame operator that has weight nearly 2 on a chosen infinite set and a small weight a_k on its complement.

The chosen infinite sets partition the coordinates. Thus, when all normalized bands are united, every coordinate receives one large weight and the sum of all remaining small weights. After scaling the kth band by r_k, the common small weights still sum to a strictly positive number. This simultaneously makes the original sequence and its normalization frames, while geometrically separated norms make the bands exact.

3 Counterexample theorem

For a subset $E \subseteq \mathbb{N}$, write P_E for the orthogonal projection onto $\overline{\operatorname{span}}\{e_j : j \in E\}$.

Theorem 1. *There exists a normalizable frame in $\ell^2(\mathbb{N})$ belonging to category (c) of Theorem 3.13. More precisely, it can be chosen so that its normalized sequence has optimal frame bounds 2 and 3, the original sequence has optimal frame bounds $1/7$ and $11/28$, and its norm bands have optimal bounds*

$$A_k = 2^{-3k}, \qquad B_k = 4^{-k}(2 - 2^{-k}).$$

Consequently Conjecture 3.14 is false.

Proof. Let $\mathcal{H} = \ell^2(\mathbb{N})$ with its standard orthonormal basis $(e_j)_{j\geq 1}$. For instance, take the explicit partition

$$E_k = \{2^{k-1}(2m-1) : m \geq 1\}, \qquad \mathbb{N} = \bigsqcup_{k\geq 1} E_k.$$

Every E_k is infinite. Both E_k and E_k^c are countably infinite, so let $\sigma_k : E_k \to E_k^c$ match their increasing enumerations. Put

$$a_k = 2^{-k}, \qquad r_k = 2^{-k}.$$

For $p \in E_k$, define two unit vectors

$$u_{k,p}^{\pm} = \sqrt{1 - \frac{a_k}{2}}\, e_p \pm \sqrt{\frac{a_k}{2}}\, e_{\sigma_k(p)}, \tag{1}$$

and let $\mathcal{U}_k = (u^{\pm}_{k,p})_{p\in E_k}$. Finally set

$$x^{\pm}_{k,p} = r_k u^{\pm}_{k,p}. \tag{2}$$

This countable double family may of course be enumerated as a sequence.

We first compute the frame operator of one unit-vector block. For fixed k, p, the opposite signs in (1) cancel the cross terms:

$$u^{+}_{k,p} \otimes u^{+}_{k,p} + u^{-}_{k,p} \otimes u^{-}_{k,p} = (2 - a_k)e_p \otimes e_p + a_k e_{\sigma_k(p)} \otimes e_{\sigma_k(p)}. \tag{3}$$

For distinct $p \in E_k$, the pairs $\{p, \sigma_k(p)\}$ are disjoint. Since σ_k maps E_k onto its complement, summing (3) gives the exact block operator

$$S_k = a_k I + (2 - 2a_k)P_{E_k}. \tag{4}$$

Thus $\mathcal{U}_k$ is a frame for $\mathcal{H}$ with optimal lower and upper bounds a_k and $2 - a_k$.

The normalization of the sequence in (2) is precisely $\mathcal{U} = \bigcup_{k\geq 1} \mathcal{U}_k$. Let $j \in E_m$. Formula (4) and $\sum_{k\geq 1} a_k = 1$ yield

$$\left(\sum_{k\geq 1} S_k\right) e_j = \left(2 - a_m + \sum_{k\neq m} a_k\right) e_j = (3 - 2a_m)e_j. \tag{5}$$

These diagonal values begin at 2 and increase to 3. Hence $\mathcal{U}$ is a frame with optimal bounds 2 and 3.

Next consider the original scaled sequence $X = (x^{\pm}_{k,p})_{k,p,\pm}$. Its frame operator is

$$T = \sum_{k\geq 1} r_k^2 S_k.$$

The common diagonal contribution is

$$C := \sum_{k\geq 1} r_k^2 a_k = \sum_{k\geq 1} 2^{-3k} = \frac{1}{7}. \tag{6}$$

For $j \in E_m$, equations (4) and (6) give

$$Te_j = \big(C + r_m^2(2 - 2a_m)\big)\, e_j. \tag{7}$$

The second term is nonnegative and tends to zero. It is maximal at $m = 1$, where it equals $1/4$. The diagonal values in (7) therefore have infimum $1/7$ and maximum $1/7 + 1/4 = 11/28$. Thus X is a frame with precisely those optimal bounds. Together with (5), this proves that X is a normalizable frame.

It remains to identify its norm bands. All vectors in the kth block have norm r_k. Define

$$\delta_k = 3\, 2^{-(k+1)} = \frac{3}{2} r_k. \tag{8}$$

Then $\delta_k \downarrow 0$ and

$$\delta_{k+1} = \frac{3}{4} r_k < r_k < \frac{3}{2} r_k = \delta_k.$$

The neighboring norms satisfy $r_{k-1} = 2r_k > \delta_k$ (when $k > 1$) and $r_{k+1} = r_k/2 < \delta_{k+1}$; all more distant norms are farther away. Consequently

$$\{x \in X : \delta_{k+1} \leq \|x\| < \delta_k\} = r_k \mathcal{U}_k. \tag{9}$$

By (4), this band is a frame with optimal bounds

$$A_k = r_k^2 a_k = 2^{-3k}, \qquad B_k = r_k^2(2 - a_k) = 4^{-k}(2 - 2^{-k}).$$

In particular, $\sum_k A_k = 1/7$ and $B_k \to 0$. Every condition in category (c) is satisfied, completing the counterexample. □

4 Verification and scope

The proof is exact and uses no numerical lemma. All relevant operators are positive and diagonal in the standard basis. The series may therefore be checked coordinatewise, while the uniform bounds in (5) and (7) ensure that the resulting quadratic forms define bounded frame operators.

The counterexample gives a full negative answer to Conjecture 3.14. It does not classify normalizable frames, nor does it address other open questions in the source paper.

5 Novelty check and review recommendation

The local run indexes were searched for arXiv:2308.13071, the exact title, and the category-(c) conjecture, with no duplicate result found. Bounded web searches on August 9, 2026 used the exact conjecture sentence, the exact title together with "category (c)," and the title/author together with "conjecture." They found the arXiv paper, its 2024 journal publication, an author thesis, and unrelated later citations, but no resolution of Conjecture 3.14 or this diagonal-block construction. The result's novelty is therefore plausible but remains subject to expert literature review.

Human review is strongly recommended. The decisive checks are (3), the three diagonal calculations (5)–(7), and the exact band isolation in (9).

References

[1] P.-T. Yu, *Frame-normalizable sequences*, Adv. Comput. Math. **50** (2024), Paper 89; arXiv:2308.13071; doi:10.1007/s10444-024-10182-z.